MySQL

1:

DESCRIBE OrderLine_T;

Query #1 Execution time: 16.2ms

Field	Type	Null	Key	Default	Extra
OrderID	int	NO	PRI	null	
ProductID	int	NO	PRI	null	
OrderedQuantity	int	YES		null	

2:

SELECT ProductDescription, ProductFinish, ProductStandardPrice

FROM Product_T

WHERE ProductFinish IN ('Natural Ash', 'White Ash')

AND ProductDescription LIKE '%Desk%';

Query #1 Execution time: 0.25ms

ProductDescription	ProductFinish	ProductStandardPrice
Computer Desk	Natural Ash	375.0
8-Drawer Desk	White Ash	750.0

3:

SELECT OrderID, OrderDate, CustomerID

FROM Order_T

WHERE CustomerID IN (1, 12)

AND OrderDate BETWEEN '2023-01-01' AND '2023-12-31';

Query #1 Execution time: 0.31ms

OrderID	OrderDate	CustomerID
1011	2023-09-14	1
1015	2023-10-08	1
1013	2023-10-05	12
1017	2023-10-12	12

4:

```
SELECT ProductDescription,
    ProductStandardPrice,
    (ProductStandardPrice - Saleprice)
      / ProductStandardPrice * 100 AS PercentOff
FROM Product_T
ORDER BY PercentOff DESC;
```

Query #1 Execution time: 0.27ms

ProductDescription	ProductStandardPrice	PercentOff
Coffee Table	200.0	36.25
End Table	175.0	15.0
Computer Desk	375.0	15.0
Writers Desk	325.0	15.0
Computer Desk	250.0	15.0
Entertainment Center	650.0	10.0
8-Drawer Desk	750.0	10.0
Dining Table	800.0	10.0

5:

```
SELECT DISTINCT ProductLineID, ProductFinish
FROM Product_T
ORDER BY ProductLineID, ProductFinish;
```

Query #1 Execution time: 0.42ms

ProductLineID	ProductFinish
1	Cherry
2	Natural Ash
2	White Ash
3	Natural Maple
3	Walnut

6:

```
SELECT CustomerID,
    COUNT(*) AS NumOrders,
    MIN(OrderDate) AS FirstOrder
FROM Order_T
GROUP BY CustomerID
ORDER BY NumOrders DESC;
```

Query #1 Execution time: 15.61ms

CustomerID	NumOrders	FirstOrder
1	7	2022-08-10
12	5	2022-08-13
15	4	2022-08-16
2	3	2023-09-21
3	3	2022-08-14
4	2	2022-08-12
5	2	2023-10-10
8	2	2023-10-07
11	2	2023-10-14

7:

```
SELECT ProductFinish,
    ROUND(AVG(ProductStandardPrice), 2) AS AvgPrice
FROM Product_T
GROUP BY ProductFinish
HAVING AVG(ProductStandardPrice) > 300
ORDER BY AvgPrice DESC;
```

Query #1 Execution time: 0.56ms

ProductFinish	AvgPrice
White Ash	750.0
Natural Maple	650.0
Natural Ash	458.33

8:

```
SELECT P.ProductDescription, P.ProductStandardPrice
FROM Product_T P, ProductLine_T PL
WHERE P.ProductLineID = PL.ProductLineID
 AND PL.ProductLineName = 'Scandinavia';
```

Query #1 Execution time: 0.29ms

ProductDescription	ProductStandardPrice
Coffee Table	200.0
Computer Desk	375.0
8-Drawer Desk	750.0
Dining Table	800.0

9:

```
SELECT P.ProductDescription,
    OL.OrderedQuantity,
    P.ProductStandardPrice
FROM OrderLine_T OL
 INNER JOIN Product_T P
  ON OL.ProductID = P.ProductID
WHERE OL.OrderID = 1026;
```

Query #1 Execution time: 1.55ms

ProductDescription	OrderedQuantity	ProductStandardPrice
Coffee Table	5	200.0
Entertainment Center	5	650.0
Computer Desk	9	250.0

10:

```
SELECT T.TerritoryName,
    COUNT(DB.CustomerID) AS NumCustomers
FROM Territory_T T
 LEFT JOIN DoesBusinessIn_T DB
   ON T.TerritoryID = DB.TerritoryID
GROUP BY T.TerritoryID, T.TerritoryName
ORDER BY NumCustomers DESC;
```

Query #1 Execution time: 0.55ms

TerritoryName	NumCustomers
SouthWest	3
NorthEast	2
SouthEast	1
Central	1
NorthWest	0

11:

```
SELECT O.OrderID,
    O.OrderDate,
    P.ProductDescription,
    OL.OrderedQuantity,
    OL.OrderedQuantity * P.ProductStandardPrice AS LineTotal
```

```
FROM Customer_T C
  INNER JOIN Order_T O
    ON C.CustomerID = O.CustomerID
  INNER JOIN OrderLine_T OL
    ON O.OrderID = OL.OrderID
  INNER JOIN Product_T P
    ON OL.ProductID = P.ProductID
WHERE C.CustomerName = 'Contemporary Casuals'
  AND O.OrderDate BETWEEN '2023-01-01' AND '2023-12-31'
ORDER BY O.OrderID;
```

Query #1 **Execution time: 0.58ms**

OrderID	OrderDate	ProductDescription	OrderedQuantity	LineTotal
1011	2023-09-14	Coffee Table	1	200.0
1011	2023-09-14	Entertainment Center	2	1300.0
1011	2023-09-14	Writers Desk	2	650.0
1015	2023-10-08	Computer Desk	8	2000.0

12:

```
SELECT PL.ProductLineName,
     SUM(OL.OrderedQuantity * P.ProductStandardPrice) AS SalesValue
FROM ProductLine_T PL
  INNER JOIN Product_T P
    ON PL.ProductLineID = P.ProductLineID
  INNER JOIN OrderLine_T OL
    ON P.ProductID = OL.ProductID
GROUP BY PL.ProductLineID, PL.ProductLineName
ORDER BY SalesValue DESC;
```

Query #1 Execution time: 0.55ms

ProductLineName	SalesValue
Scandinavia	39100.0
Country Look	22800.0
Cherry Tree	9550.0

13:

```
SELECT A.ProductDescription AS Product1,
      B.ProductDescription AS Product2,
      A.ProductFinish
FROM Product_T A
 INNER JOIN Product_T B
   ON A.ProductFinish = B.ProductFinish
  AND A.ProductID < B.ProductID;
```

Query #1 Execution time: 0.54ms

Product1	Product2	ProductFinish
End Table	Writers Desk	Cherry
Coffee Table	Computer Desk	Natural Ash
Coffee Table	Dining Table	Natural Ash
Computer Desk	Dining Table	Natural Ash

14:

```
SELECT ProductDescription, ProductFinish
FROM Product_T
WHERE ProductLineID =
     (SELECT ProductLineID
      FROM Product_T
```

```
    WHERE ProductDescription = 'Dining Table');
```

Query #1 **Execution time: 0.46ms**

ProductDescription	ProductFinish
Coffee Table	Natural Ash
Computer Desk	Natural Ash
8-Drawer Desk	White Ash
Dining Table	Natural Ash

15:

```
SELECT ProductDescription, ProductFinish
FROM Product_T
WHERE ProductID IN
    (SELECT ProductID
     FROM OrderLine_T
     WHERE OrderedQuantity >= 8);
```

Query #1 **Execution time: 0.41ms**

ProductDescription	ProductFinish
End Table	Cherry
Coffee Table	Natural Ash
Computer Desk	Natural Ash
Computer Desk	Walnut

16:

```
SELECT P.ProductID, P.ProductDescription
FROM Product_T P
WHERE NOT EXISTS
    (SELECT *
```

```
FROM Order_T O
  INNER JOIN OrderLine_T OL
    ON O.OrderID = OL.OrderID
WHERE O.CustomerID = 2
  AND OL.ProductID = P.ProductID);
```

Query #1 **Execution time: 0.48ms**

ProductID	ProductDescription
5	Writers Desk
6	8-Drawer Desk